

The length of the link is the distance between the centres of two consecutive middle rings. The length of the chain is, outside of one handle to the outside of the other handle.

The following are the various types of chains, in common use,

- (i) Metric chain.
- (ii) Gunter's chain or surveyor's chain.
- (iii) Engineer's chain.
- (iv) Revenue chain.
- (v) Steel band or band chain.

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1. **Metric Chain :** Metric chains are available in lengths of 5, 10, 20 and 30 metres. Among the 20 m and 30 m are commonly used. 20 m chain is divided into 100 links and 30 m chain is divided into 150 links. Length of each link is 0.2 m. The brass tallies are fixed at every 5 m length and small brass rings are provided at every metre length. The least count of the metric chain is 0.2 m or 20 cm. (Fig. 2.7).

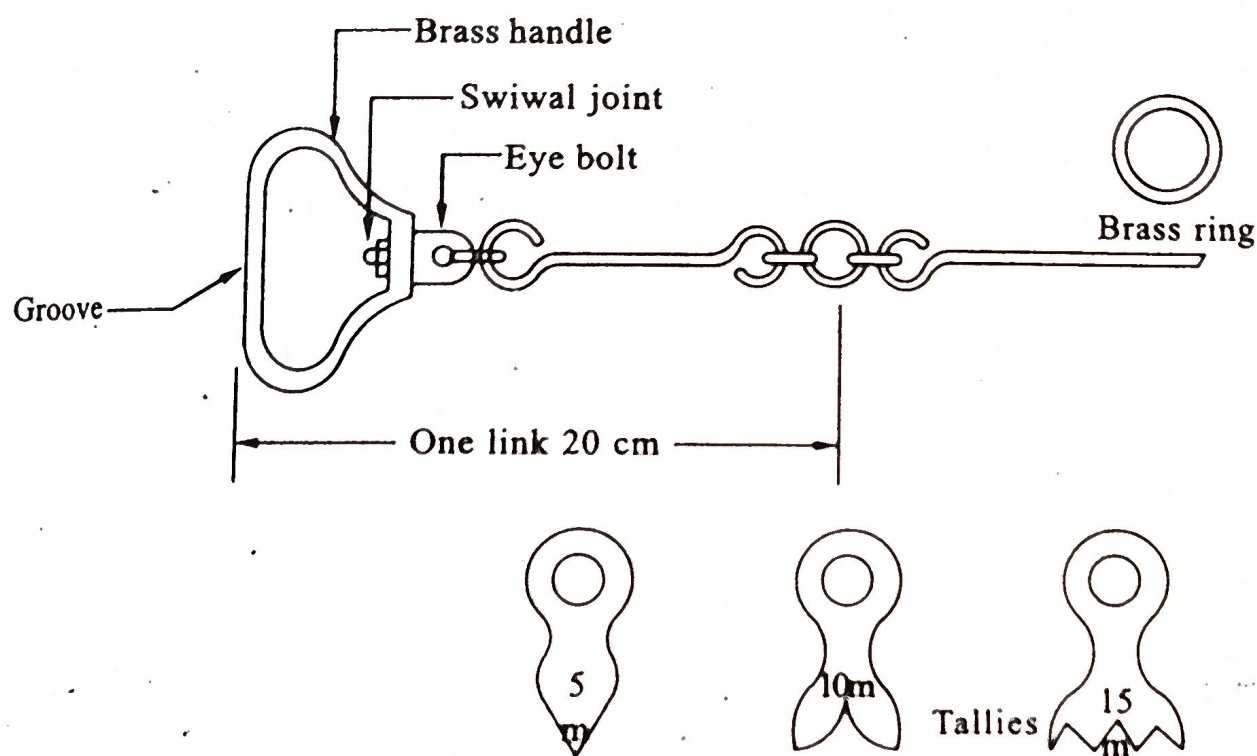


FIG 2.7 :

2. **Gunter's Chain or Surveyor's Chain :** This chain is 66 feet long and having 100 links. So, each is of 0.66 ft. It is very useful for measuring the distances in miles and furlongs. It is also useful for measuring the area in acres.

1 Gunter's chain = 22 yards

10 Gunter's chains = 1 furlong

80 Gunter's chains = 1 mile

1 Square gunter chain = 484 Sq. yards (10 cents).

10 Square gunter chains = 1 Acre = 4840 Sq.yards (100 cents)

3. **Engineer's Chain :** The engineer's chain is 100 ft long and divided into 100 links. So, length of each link is of 1 foot. Tallies or tags are provided at every 10 links (10 ft.), the central tally being round. Engineer's chain was previously used for all engineering works.

4. **Revenue Chain** : This chain is 33 *ft.* long and divided into 16 links. Length of each link being 2 *ft* 0.75 inches. It is commonly used for measuring fields in cadestral surveys and also used by the revenue department for measuring land in bighas. One square Revenue chain equal to one bigha.
5. **Steel Band or Band Chain** : It consists of a ribbon of steel of width 16 *mm* and of length 20 or 30 *m*. It has a brass handle at both ends. On one side it is graduated in metres, decimetres and centimetres and on the other side graduation are provided at 0.2 *m*. For carrying and handling it is wound on special steel crosses. The steel band should be used for accurate work. It should be operated by skilled chainmen.

ARROWS

Arrows are made of good quality hardened